Memo – WTW158 – Sem 2 – 2010

AFDELING A : MEERVOUDIGE KEUSEVRAE - 15 PUNTE

**DOEN HIERDIE AFDELING EERSTE, AANGESIEN U DIE MERKLEESVORM MET U ANTWOORDE NA EEN UUR MOET INGEE.**

Gebruik 'n sagte potlood.

Vul jou persoonlike inligting in potlood in op die merkleesvorm se **KANT 2**.

Vul jou studentenommer in van bo na onder en kodeer dit.

**Indien jou studentenommer verkeerd gekodeer is of nie ingevul is nie, sal jy nie punte vir afdeling A kry nie.**

Beantwoord vrae 1 tot 15 op die MERKLEESVORM se **KANT 2**.

**Indien kant 1 gebruik word sal dit nie nagesien word nie.**

Jy mag nie verkeerde antwoorde uitvee op die merkleesvorm nie, dus omkring jou antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as jy seker is van jou keuse.

SECTION A: MULTIPLE CHOICE QUESTIONS - 15 MARKS

**DO THIS SECTION FIRST, AS YOU HAVE TO HAND IN THE OPTIC READER FORM WITH YOUR ANSWERS AFTER ONE HOUR.**

Use a soft pencil.

Fill in your personal information in pencil on the optic reader form on **SIDE 2**.

Fill in your student number from top to bottom and then code it.

**If you make a mistake when coding your student number or if the student number is not coded, you will get no marks for section A.**

Answer questions 1 to 15 on the OPTIC READER FORM on **SIDE 2**.

**If side 1 is used it will not be marked.**

As you may not erase wrong answers on the optic reader form, first circle your answers in this paper and only mark the answers on the form when you are certain of your choice.

memo, Short questions at the back

**Vraag 1 / Question 1**

$$\lim_{x\to -3} \sqrt{1-x}$$

| | | | |
|---|---|---|---|
| [1a] $= 2$ | [1b] $= \pm 2$ | [1c] bestaan nie / does not exist | [1d] Geen van hierdie / None of these |

**Vraag 2 / Question 2**

$$\lim_{x\to\infty} \frac{4x^2-3x+2}{1-3x-4x^2}$$

| | | | | |
|---|---|---|---|---|
| [2a] $= \infty$ | [2b] $= -\infty$ | [2c] $= 4$ | [2d] $= -1$ | [2e] Geen van hierdie / None of these |

**Vraag 3 / Question 3**

$$\lim_{x\to\infty} (\sqrt{x}-x)$$

| | | | |
|---|---|---|---|
| [3a] $= 0$ | [3b] $= \infty$ | [3c] $= -\infty$ | [3d] bestaan nie / does not exist |
| [3e] Geen van hierdie / None of these | | | |

**Vraag 4 / Question 4**

$$\lim_{x\to 3^-} \frac{-2x}{x^2-9}$$

| | | | | | |
|---|---|---|---|---|---|
| [4a] $= \infty$ | [4b] $= -\infty$ | [4c] $= -6$ | [4d] $= 0$ | [4e] $= \frac{-2}{3}$ | [4f] Geen van hierdie / None of these |

## Vraag 5 / Question 5

$$\lim_{x\to\infty} \frac{\sqrt{x-x^3}}{x}$$

| | | | | |
|---|---|---|---|---|
| [5a] $=\infty$ | [5b] $=-\infty$ | [5c] $=0$ | [5d] $=1$ | [5e] $=\lim_{x\to\infty}\sqrt{1-x^2}$ |
| [5f] bestaan nie aangesien $\lim_{x\to\infty}\sqrt{x-x^3}$ nie gedefinieer is nie<br>does not exist because $\lim_{x\to\infty}\sqrt{x-x^3}$ is not defined | | | | |
| [5g] Geen van hierdie / None of these | | | | |

## Vraag 6 / Question 6

Laat / Let $f(x) = \ln(1+x)$

Die grootste interval waarop $f$ kontinu is, is
The largest interval on which $f$ is continuous is

| | | | |
|---|---|---|---|
| [6a] $(-\infty,\infty)$ | [6b] $[0,\infty)$ | [6c] $[-1,\infty)$ | [6d] $(-1,\infty)$ |
| [6e] $(0,\infty)$ | [6f] Geen van hierdie / None of these | | |

## Vraag 7 / Question 7

Laat / Let $f(x) = x^2+\sqrt{x}$ met / with $f'(x) = 2x+\frac{1}{2\sqrt{x}}$.

Die grootste interval waarop $f$ differensieerbaar is, is
The largest interval on which $f$ is differentiable is

| | | | |
|---|---|---|---|
| [7a] $(-\infty,0)\cup(0,\infty)$ | [7b] $[0,\infty)$ | [7c] $(0,\infty)$ | [7d] Geen van hierdie / None of these |

## Vraag 8 / Question 8

Die vergelyking van die raaklyn aan die kromme $y = x^3$ in die punt waar $x=-1$ is
The equation of the tangent line to the curve $y=x^3$ at the point where $x=-1$ is

| | | | |
|---|---|---|---|
| [8a] $y-1=3x^2(x-1)$ | [8b] $y+1=3x^2(x+1)$ | [8c] $y+1=3(x-1)$ | [8d] $y+1=3(x+1)$ |
| [8e] $y-1=3(x+1)$ | [8f] $y-1=3(x-1)$ | [8g] Geen van hierdie / None of these | |

## Vraag 9 / Question 9

$\lim_{h\to 0}\frac{4\times 2^h-4}{h}$ is die afgeleide van die funksie

$\lim_{h\to 0}\frac{4\times 2^h-4}{h}$ is the derivative of the function

| | | | |
|---|---|---|---|
| [9a] $f(x)=2^x$ | [9b] $f(x)=4\times 2^x$ | [9c] $f(x)=2^x$ in / at $x=0$ | [9d] $f(x)=4\times 2^x$ in / at $x=0$ |
| [9e] Geen van hierdie / None of these | | | |

**Vraag 10 / Question 10**

Beskou die funksie $f(x) = \frac{-x}{x^2 - 1}$. Watter bewering hieronder is waar?

Consider the function $f(x) = \frac{-x}{x^2 - 1}$. Which statement below is true?

| | |
|---|---|
| [10a] | Die funksie het geen horisontale asimptote nie.<br>The function has no horizontal asymptotes. |
| [10b] | Die funksie het 'n horisontale asimptoot by $x = 0$.<br>The function has a horizontal asymptote at $x = 0$. |
| [10c] | Die funksie het 'n horisontale asimptoot by $y = 0$.<br>The function has a horizontal asymptote at $y = 0$. |
| [10d] | Die funksie het horisontale asimptote by $x = 1$ en $x = -1$.<br>The function has horizontal asymptotes at $x = 1$ and $x = -1$. |
| [10e] | Geen van hierdie / None of these |

**Vraag 11 / Question 11**

Beskou die funksie $f(x) = \frac{1 - x}{x^2 - 1}$. Watter bewering hieronder is waar?

Consider the function $f(x) = \frac{1 - x}{x^2 - 1}$. Which statement below is true?

| | |
|---|---|
| [11a] | Die funksie het geen vertikale asimptote nie.<br>The function has no vertical asymptotes. |
| [11b] | Die funksie het vertikale asimptote by $x = 1$ en $x = -1$.<br>The function has vertical asymptotes at $x = 1$ and $x = -1$. |
| [11c] | Die funksie het slegs 'n vertikale asimptoot by $x = -1$.<br>The function has only a vertical asymptote at $x = -1$. |
| [11d] | Die funksie het slegs'n vertikale asimptoot by $x = 1$.<br>The function has a only vertical asymptote at $x = 1$. |
| [11e] | Geen van hierdie / None of these |

**Vraag 12 / Question 12**

Gestel $f$ is 'n funksie met $\lim_{x \to 5} \frac{f(x) - f(5)}{x - 5} = 0$. Beskou die bewerings hieronder:

Assume that $f$ is a function such that $\lim_{x \to 5} \frac{f(x) - f(5)}{x - 5} = 0$. Consider the statements below:

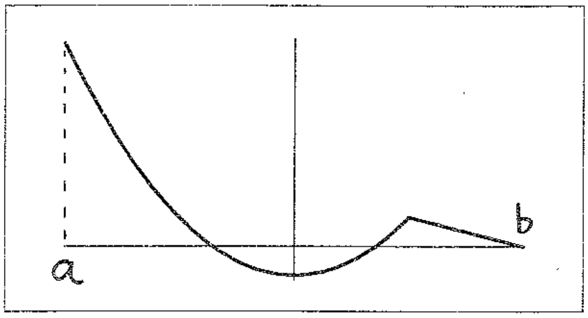
**I** $f$ is kontinu in $x = 5$ / $f$ is continuous at $x = 5$

**II** $f$ is differensieerbaar in $x = 5$ / $f$ is differentiable at $x = 5$

Watter bewering(s) is waar? / Which statement(s) is(are) true?

| | | |
|---|---|---|
| [12a] slegs / only **I** | [12b] slegs / only **II** | [12c] **I** en / and **II** |
| [12d] Geen van hierdie / None of these | | |

**Gebruik die grafiek van die funksie $f$ hieronder om Vrae 13 en 14 te beantwoord.**
**Use the graph of the function $f$ below to answer Questions 13 and 14.**




Funksie / Function $f$

## Vraag 13 / Question 13

Beskou die bewerings hieronder: / Consider the statements below:

**I** $f$ is kontinu op sy definisieversameling / $f$ is continuous on its domain

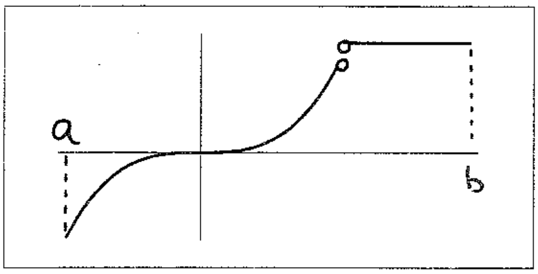
**II** $f$ is differensieerbaar op sy definisieversameling / $f$ is differentiable on its domain

Watter bewering is waar? / Which statement is true?

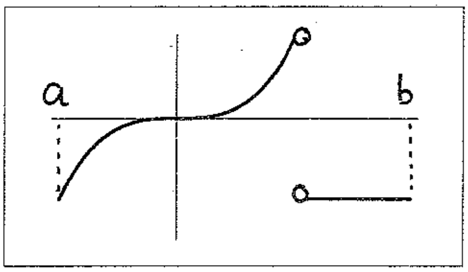
| | | |
|---|---|---|
| [13a] slegs / only **I** | [13b] slegs / only **II** | [13c] **I** en / and **II** |
| [13d] Geen van hierdie / None of these | | |

## Vraag 14 / Question 14

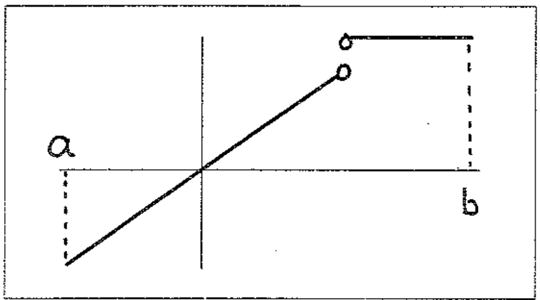
Watter van die grafieke hieronder kan moontlik die afgeleide funksie van $f$ wees?
Which of the graphs below can possibly be the derivative of the function $f$?



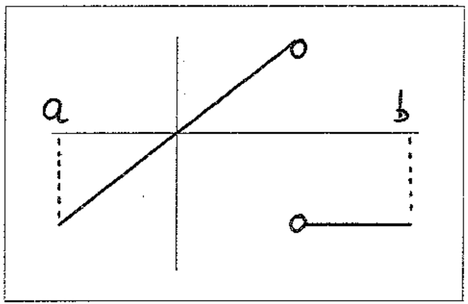

**I**




**II**




**III**




**IV**

| | | |
|---|---|---|
| [14a] slegs / only **I** | [14b] slegs / only **III** | [14c] **I** en / and **III** |
| [14d] slegs / only **II** | [14e] slegs / only **IV** | [14f] **II** en / and **IV** |

**Vraag 15 / Question 15**

Indien $f(x) = \sin^2(x)$, dan is $f'(\frac{\pi}{3})$ = / If $f(x) = \sin^2(x)$, then $f'(\frac{\pi}{3})$ =

| | | | | |
|---|---|---|---|---|
| [15a] $= \frac{3}{4}$ | [15b] $= \frac{1}{4}$ | [15c] $= 1$ | [15d] $= \frac{\sqrt{3}}{2}$ | [15e] $= \sqrt{3}$ |
| [15f] Geen van hierdie / None of these | | | | |

[15]

AFDELING B: MAKSIMUM 35 PUNTE

**BEANTWOORD ALLE VERDERE VRAE OP HIERDIE VRAESTEL IN INK. TOON ALLE BEREKENINGE. VERDUIDELIK U ANTWOORDE.**

SECTION B: MAXIMUM 35 MARKS

**ANSWER ALL THE FOLLOWING QUESTIONS ON THIS PAPER IN INK. SHOW ALL CALCULATIONS AND EXPLAIN YOUR ANSWERS.**

**Vraag 16 / Question 16**

Bepaal die volgende limiete. Toon die berekeninge.

Find the following limits. Show the calculations.

**i** $\lim_{x\to 0} \frac{\sin(2x^2)}{x}$ (form $\frac{0}{0}$)

$= \lim_{x\to 0} \frac{\sin(2x^2)}{2x^2} \times 2x$

$= 1 \times 2 \times 0 = 0$

[1]

**ii** $\lim_{x\to 2^-} \frac{2-x}{x^2-4}$ (form $\frac{0}{0}$)

$= \lim_{x\to 2^-} \frac{-(x-2)}{(x-2)(x+2)}$

[2]

$= \lim_{x\to 2^-} \frac{-1}{x+2}$

$= \frac{-1}{4}$

**iii** $\lim_{x \to 1} \frac{x^2 - 2x + 1}{\sqrt{x} - 1}$ (form $\frac{0}{0}$)

$= \lim_{x \to 1} \frac{x^2 - 2x + 1}{\sqrt{x} - 1} \times \frac{\sqrt{x} + 1}{\sqrt{x} + 1}$

$= \lim_{x \to 1} \frac{(x^2 - 2x + 1)(\sqrt{x} + 1)}{x - 1}$

$= \lim_{x \to 1} \frac{(x-1)(x-1)(\sqrt{x} + 1)}{x - 1}$

$= \lim_{x \to 1} (x-1)(\sqrt{x} + 1) = 0 \times 2 = 0$

or $\lim_{x \to 1} \frac{x^2 - 2x + 1}{\sqrt{x} - 1}$

$= \lim_{x \to 1} \frac{(x-1)(x-1)}{\sqrt{x} - 1}$

$= \lim_{x \to 1} \frac{(\sqrt{x}-1)(\sqrt{x}+1)(x-1)}{\sqrt{x} - 1}$

$= \lim_{x \to 1} (\sqrt{x}+1)(x-1)$

$= 2 \times 0 = 0$

[2]

**iv** $\lim_{x \to -\infty} \frac{\sqrt{16x^2 + 9x - 10}}{8x - 7}$ (form $\frac{\infty}{\infty}$)

$= \lim_{x \to -\infty} \frac{\sqrt{x^2(16 + 9/x - 10/x^2)}}{x(8 - 7/x)}$

$= \lim_{x \to -\infty} \frac{|x|\sqrt{16 + 9/x - 10/x^2}}{x(8 - 7/x)}$ as $\sqrt{x^2} = |x|$

[2]

$= \lim_{x \to -\infty} \frac{-x\sqrt{16 + 9/x - 10/x^2}}{x(8 - 7/x)}$ as $x$ is negative, $|x| = -x$.

$= \lim_{x \to -\infty} - \frac{\sqrt{16 + 9/x - 10/x^2}}{8 - 7/x}$

$= - \frac{\sqrt{16}}{8} = - \frac{4}{8} = - \frac{1}{2}$

**Vraag 17 / Question 17**

Beskou die funksie $f$: / Consider the function $f$:

$$f(x) = \begin{cases} a|x-3| & \text{as / if } x > 2 \\ 4 & \text{as / if } x = 2 \\ -x^2 + 4x - 4 + a & \text{as / if } x < 2 \end{cases}$$

Bepaal die getal $a$ sodat die funksie $f$ kontinu in $x = 2$ sal wees.
Toon DUIDELIK aan hoe u die definisie van kontinuiteit gebruik.

Find the number $a$ such that $f$ will be continuous at $x = 2$.
CLEARLY show how you use the definition of continuity.

(i) $f(2) = 4$, exists for all $a \in \mathbb{R}$

(ii) $\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} a|x-3| = a|-1| = a$

$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} (-x^2+4x-4+a) = -4+8-4+a = a$

$\lim_{x\to 2} f(x)$ exists $\iff \lim_{x\to 2^+} f(x) = \lim_{x\to 2^-} f(x) \iff a = a$

$\lim_{x\to 2} f(x)$ exists for all $a \in \mathbb{R}$

(iii) $f(2) = \lim_{x\to a} f(a) \Rightarrow 4 = a$. continuous if $a = 4$.

[4]

**Vraag 18 / Question 18**

Bepaal die afgeleides van die volgende funksies.
**MOENIE U ANTWOORD VEREENVOUDIG NIE.**

Find the derivatives of the following functions.
**DO NOT SIMPLIFY YOUR ANSWER.**

**i** $f(x) = \frac{10}{x} + 10^x$

$f'(x) = \frac{-10}{x^2} + (\ln 10)10^x$

[2]

**ii** $f(t) = e^{2t}\cos(t^7 - 3)$

$f'(t) = 2e^{2t}\cos(t^7-3) + e^{2t}(-\sin(t^7-3)) \times 7t^6$

[2]

**iii** $y = 3^{\sec 4x}$

$$\frac{dy}{dx} = (\ln 3)\ 3^{\sec 4x} \times \sec 4x \tan 4x \times 4$$

[1]

**iv** $f(z) = \ln(z^3 - z)$

$$f'(z) = \frac{3z^2 - 1}{z^3 - z}$$

[1]

**v** $f(x) = \tan^2(x^3 + e)$

$$f'(x) = 2\tan(x^3 + e) \times \sec^2(x^3 + e) \times 3x^2$$

[1]

**vi** $y = \sin(4\sqrt{x})$

$$\frac{dy}{dx} = \cos(4\sqrt{x}) \times 4 \times \frac{1}{2}x^{-\frac{1}{2}}$$

[1]

**vii** $f(t) = \sqrt{1 + \sin^{-1}(t)}$ (Notasie: $\sin^{-1}$ = bgsin / Notation: $\sin^{-1}$ = arcsin)

$$f'(t) = \frac{1}{2}(1 + \sin^{-1}(t))^{-\frac{1}{2}} \times \frac{1}{\sqrt{1 - t^2}}$$

[1]

**viii** $y = (1 - x)^{\sin x^3}$

$$\ln y = \ln(1-x)^{\sin x^3} = \sin x^3 \ln(1-x)$$

$$\frac{1}{y}\frac{dy}{dx} = \cos x^3 \times 3x^2 \times \ln(1-x) + \sin x^3 \frac{-1}{1-x}$$

$$\frac{dy}{dx} = y\left[\cos x^3 \times 3x^2 \times \ln(1-x) + \sin x^3 \times \frac{-1}{1-x}\right]$$

[3]

**Vraag 19 / Question 19**

i Beskou die kromme $x^2y^3 + 6x = y + 12$. / Consider the curve $x^2y^3 + 6x = y + 12$.

Bepaal $\frac{dy}{dx}$. / Find $\frac{dy}{dx}$.

$$2xy^3 + x^2 \times 3y^2 \frac{dy}{dx} + 6 = \frac{dy}{dx}$$

$$\therefore 3x^2y^2\frac{dy}{dx} - \frac{dy}{dx} = -2xy^3 - 6$$

$$\frac{dy}{dx} = \frac{-2xy^3 - 6}{3x^2y^2 - 1}$$

[2]

ii Bepaal die vergelyking van die raaklyn aan die kromme in die punt $(1,2)$.

Find the equation of the tangent line to the curve in the point $(1,2)$

$$m = \left.\frac{dy}{dx}\right|_{\substack{x=1\\y=2}} = \frac{-2\times 8 - 6}{3\times 4 - 1} = \frac{-22}{11} = -2$$

$$m = \frac{y-y_1}{x-x_1} \Rightarrow -2 = \frac{y-2}{x-1} \Rightarrow y-2 = -2(x-1)$$

$$\Rightarrow y = -2x + 4$$

[2]

**Vraag 20 / Question 20**

Bewys dat $\frac{d}{dx}\cot 3x = -3\operatorname{cosec}^2(3x)$ deur die kwosiëntreël en die afgeleides van $y = \sin 3x$ en $y = \cos 3x$ te gebruik.

Prove that $\frac{d}{dx}\cot 3x = -3\operatorname{cosec}^2(3x)$ by using the quotient rule and the derivatives of $y = \sin 3x$ and $y = \cos 3x$.

$$\frac{d}{dx}\cot 3x = \frac{d}{dx}\frac{\cos 3x}{\sin 3x}$$

$$= \frac{\left[\frac{d}{dx}\cos 3x\right]\sin 3x - \cos 3x\left[\frac{d}{dx}\sin 3x\right]}{\sin^2 3x}$$ quotient rule

$$= \frac{-3\sin 3x.\sin 3x - \cos 3x \times 3\cos 3x}{\sin^2 3x}$$

[2]

$$= -3\left[\sin^2 3x + \cos^2 3x\right]/\sin^2 3x = \frac{-3}{\sin^2 3x} = -3\operatorname{cosec}^2 3x$$

**Vraag 21 / Question 21**

**i** Gebruik die limiet $\lim_{h\to 0} \frac{\sin h}{h} = 1$ om $\lim_{h\to 0} \frac{\cos h - 1}{h}$ te bepaal.

Use the limit $\lim_{h\to 0} \frac{\sin h}{h} = 1$ to find $\lim_{h\to 0} \frac{\cos h - 1}{h}$.

$$\lim_{h\to 0} \frac{\cosh - 1}{h} \quad (\text{form } \tfrac{0}{0})$$

$$= \lim_{h\to 0} \frac{\cos h - 1}{h} \times \frac{\cos h + 1}{\cos h + 1}$$

$$= \lim_{h\to 0} \frac{\cos^2 h - 1}{h(\cos h + 1)} = \lim_{h\to 0} \frac{-\sin^2 h}{h(\cos h + 1)} \quad , \sin^2 h + \cos^2 h = 1$$

$$= \lim_{h\to 0} -\sin h \times \frac{\sin h}{h} \times \frac{1}{\cos h + 1} = -0 \times 1 \times \frac{1}{2} = 0$$

[2]

**ii** Gebruik die limiet-definisie van die afgeleide om te bewys dat $\frac{d}{dx}\cos x = -\sin x$.

Use the limit definition of the derivative to prove that $\frac{d}{dx}\cos x = -\sin x$.

$$\frac{d}{dx} \cos x = \lim_{h\to 0} \frac{\cos(x+h) - \cos x}{h}$$

$$= \lim_{h\to 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$$

$$= \lim_{h\to 0} \frac{\cos x(\cos h - 1) - \sin x \sin h}{h}$$

$$= \cos x \times \lim_{h\to 0} \frac{\cos h - 1}{h} - \sin x \lim_{h\to 0} \frac{\sin h}{h}$$

$$= \cos x \times 0 \quad - \sin x \times 1$$

$$= -\sin x$$

[3]

**Vraag 22 / Question 22**

Gebruik die Tussenwaardestelling om aan te toon dat daar 'n $c \in [0,4]$ is sodanig dat $c + 2\sin c - 1 = 0$. Verduidelik u antwoord.

Use the Intermediate Value Theorem to proof that there is a $c \in [0,4]$ such that $c + 2\sin c - 1 = 0$. Explain your answer.

[3]

Let $f(x) = x + 2\sin x - 1$

$f$ is continuous on $[0,4]$

$f(0) = 0 + 2\sin 0 - 1 = -1 < 0$

$f(4) = 4 + 2\sin 4 - 1$

$= 3 + 2\sin 4$

$\geq 3 - 2 = 1$ as $-1 \leq \sin x \leq 1$ for all $x$

as $f(0) < 0$ and $f(4) > 0$ and $f$ is continuous on $[0,4]$,
there must be a $c \in [0,4]$ such that

$f(c) = 0 \Rightarrow c + 2\sin c - 1 = 0$

# WTW158 Memo, short questions, semester test 2

## Question 1

$$\lim_{x\to -3}\sqrt{1-x} = \sqrt{1-(-3)} = \sqrt{4} = 2 \qquad 1(a)$$

## Question 2

$$\lim_{x\to\infty}\frac{4x^2-3x+2}{1-3x-4x^2} = \lim_{x\to\infty}\frac{4-\frac{3}{x}+\frac{2}{x^2}}{\frac{1}{x^2}-\frac{3}{x}-4} = \frac{4}{-4} = -1 \qquad 2(d)$$

## Question 3

$$\lim_{x\to\infty}(\sqrt{x}-x) \text{ (form } \infty-\infty)$$

$$= \lim_{x\to\infty}\sqrt{x}-(\sqrt{x})^2 = \lim_{x\to\infty}\sqrt{x}(1-\sqrt{x}) = -\infty \qquad 3c$$

## Question 4

$$\lim_{x\to 3^-}\frac{-2x}{x^2-9} = \lim_{x\to 3^-}\frac{-2x}{(x-3)(x+3)} = +\infty \qquad \frac{-}{-\,+} \qquad x<3 \therefore x-3<0$$

4a

## Question 5

$$\lim_{x\to\infty} x-x^3 = \lim_{x\to\infty} x(1-x^2) = \lim_{x\to\infty} x(1-x)(1+x) = -\infty$$

That means $\lim_{x\to\infty}\sqrt{x-x^3}$ is not defined $\qquad$ 5f)

## Question 6

f is continuous on its domain; $1+x>0 \Rightarrow x>-1 \qquad$ 6 d

## Question 7

The domain is $[0,\infty)$ but f is not differentiable at $x=0$

$\therefore$ Differentiable on $(0,\infty) \qquad$ 7 c

## Question 8

$f(x) = x^3 \Rightarrow f'(x) = 3x^2$ and $f(-1) = (-1)^3 = -1$ and $f'(-1) = 3$

$m = \dfrac{y - y_1}{x - x_1} \Rightarrow 3 = \dfrac{y-(-1)}{x-(-1)} \Rightarrow y+1 = 3(x+1)$ 8(a)

## Question 9

If $f(x) = 4 \times 2^x$ then $f'(0) = \lim_{h\to 0} \dfrac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \dfrac{4 \times 2^h - 4}{h}$

If $f(x) = 2^x$ then $f'(0) = \lim_{h\to 0} \dfrac{f(0+h) - f(0)}{h} = \lim_{h\to 0} \dfrac{2^h - 1}{h}$

9(a)

## Question 10

$$\left.\begin{array}{l} \lim_{x\to\infty} f(x) = \lim_{x\to\infty} \dfrac{-x}{x^2+1} = \lim_{x\to\infty} \dfrac{-1}{x+\frac{1}{x}} = 0 \\ \lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \dfrac{-1}{x+\frac{1}{x}} = 0 \end{array}\right\} \Rightarrow y = 0 \text{ is the HA}$$

10(c)

## Question 11

$f(x) = \dfrac{1-x}{x^2-1} = \dfrac{-(x-1)}{(x-1)(x+1)} = \dfrac{-1}{x+1}$

$\lim_{x\to -1^+} f(x) = -\infty$ $\lim_{x\to -1^-} f(x) = +\infty$ The only VA is $x = -1$ 11c

## Question 12

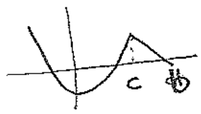
If $\lim_{x\to a} \dfrac{f(x) - f(a)}{x-a}$ exists, then $f$ is differentiable at $x = a$. If $f$ is differentiable at $x=a$ then $f$ is continuous at $x = a$. 12(c)

## Question 13

The graph of f is in "one piece", therefore f is continuous on its domain.

The graph of f has a "corner", therefore f is not differentiable in that point 13(a)

## Question 14

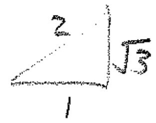
- $f'(0)=0$ (true for I, II, III, IV)
- f is decreasing on $(a,0)$ and therefore $f'$ must be negative on $(a,a)$ (true for I, II, III, IV)
- 

f is increasing on $(0,c)$ and therefore $f'$ must be positive on $(0,c)$ (true for I, II, III, IV)
- f is not differentiable at $x=c$, $f'(c)$ is not defined (true for I, II, III, IV)
- f is decreasing on $(c,b)$ therefore $f'$ is negative on $(c,b)$ true for II and IV 14(f)

## Question 15

$f(x) = \sin^2 x \Rightarrow f'(x) = 2\sin x \cos x$

$\Rightarrow f'(\frac{\pi}{3}) = 2\sin\frac{\pi}{3}\cos\frac{\pi}{3}$




$= 2\times\frac{\sqrt{3}}{2}\,\frac{1}{2} = \frac{\sqrt{3}}{2}$ 15(a)